

DESIGN FOR SOCIAL INNOVATION

Project Sprint Plan

COURSE ARC: Understand → Envision → Build → Improve → Validate → Showcase

PRIMARY DELIVERABLE: A functional proof-of-concept (POC) of the core technical component required to address the assigned problem.

Project Scope

The focus is on building the core capability with technology.

Students are not expected to build a complete production application.

- Login / authentication / registration
- User management
- Administrative dashboards
- Production application-integration plumbing beyond the core
- Fully programmed frontend

Note: The mock UI is primarily a communication and design tool. It may be created in Figma or another clickable / navigable format.

Submission Requirements - Important

The end of sprint & hackathon deliverables must be submitted as follows:

- Items Highlighted in Green are to be submitted in **Moodle**.
- Items Highlighted in Purple are to be submitted by pushing to **GitHub**.

Sprint 0 — Understand the Problem & Context

Dates : 8–12 September

Aim : Build a shared, evidence-based understanding of the assigned problem before proposing a solution.

Activities

- Analyze the project statement and clarify its scope.
- Identify users, stakeholders, setting and constraints.
- Investigate the real-world context and pain points.
- Surface assumptions that need validation.

End-of-Sprint deliverables

- **FUNCTIONAL SCOPE DOC:**
 - Problem & Context.
 - User / stakeholder map (along with what & why).
 - Key pain points and constraints (of the users).
- Open questions / assumptions to validate.

CHECKPOINT: We understand the problem we are being asked to solve.

Sprint 1 — Envision the Solution & Define the Technology

Dates : 13–26 September

Aim : Turn the problem understanding into a concrete solution concept and define the core technical component to be built.

Activities

- Study existing solutions and identify gaps.
- Ideate and select a focused solution.
- Map the intended user workflow.
- Create a clickable / navigable mock UI in Figma or equivalent.
- Decompose the solution into the core backend technical components / Services (APIs).

End-of-Sprint deliverables

- **DESIGN DOC:**
 - User context and pain-point analysis.
 - Existing-solution analysis.
 - Proposed solution and user flow.
 - Functional mock UI.
 - Core technical component definition: inputs, outputs and intended behaviour.
 - Initial technical approach (stack, frameworks/open source, models, flows, etc.).

CHECKPOINT: Every team can clearly state what they are building, the technology that will be built, and why.

Sprint 2 — Build the First Technical POC

Dates : 27 September–10 October

Aim : Move from design into implementation and establish a first working version of the core technology.

Activities

- Define technical architecture and evaluation criteria.
- Set up data, models, algorithms or processing pipeline.
- Implement the hardest technical component first.
- Establish a baseline result.
- Prepare the component for a focused Hack 1 build.

End-of-Sprint deliverables

- **TECHNICAL DESIGN DOC:**
 - Technical specification / architecture.
 - Initial implementation.
 - Baseline technical result (what worked or didn't).
- **POC v0.1 foundations ready/before Hack 1 demonstration.**

CHECKPOINT: A first working technical POC exists — even if crude or incomplete.

Hack 1 — First POC Build

Date : 10 October • Saturday • Half-day

Aim : Make the core technical component work for the first time and expose its weaknesses through a live demonstration.

Activities

- Fully focused team build:
 - Prioritize the core technical path over polish.
 - Integrate the key components into one demonstrable flow.
 - Capture failures, gaps and questions as you work.
 - Prepare a short end-of-hack demo.

End-of-Sprint deliverables

- **TECHNICAL DESIGN DOC (revised) + DEMO VIDEO.**
 - Live demonstration: input → technology → output.
 - Short explanation of what works and what does not.
 - Specific issues / questions for instructor feedback.
- **WORKING CORE POC v0.1**
 - Feedback captured to drive Sprint 3.

CHECKPOINT: Make it work. Then learn from what does not work.

Sprint 3 — Build on Hack 1, to v0.2

Dates : 11–24 October

Aim : Continue to build your solution while using the feedback and failure evidence from Hack 1 to fix the most important technical weaknesses.

Activities

- Review Hack 1 feedback and identify remaining features & priority issues.
- Investigate failure cases (in generated code) and technical bottlenecks.
- Expand / improve functionality, accuracy, speed or robustness as relevant.
- Run focused tests against defined requirements.
- Iterate on the core technology — not application plumbing.

End-of-Hackathon deliverables

- **ALL FUNCTIONS/COMPONENTS IN PLACE + TEST SPEC.**
 - Improved POC v0.2.
 - Evidence of changes made in response to feedback.
 - Technical test / evaluation results.
 - Known limitations and remaining priorities.

CHECKPOINT: The POC is materially better than the Hack 1 version.

Sprint 4 — Validate, Refine & Prepare to Complete

Dates : 25 October–7 November

Aim : Make the core technology credible, robust and ready for the final focused build in Hack 2.

Activities

- Systematically test the POC.
- Evaluate edge cases and failure modes.
- Improve robustness and reliability.
- Refine performance and output quality.
- Prepare the final demonstration and evidence.

End-of-Sprint deliverables

- **TECHNICAL DESIGN DOC (revised)**
 - Evaluation results and failure analysis.
 - Documented limitations.
- **FIRST NEAR COMPLETE POC VER 0.9**
 - Demo plan and technical evidence ready for Hack 2.

CHECKPOINT: The technology is stable enough to polish and complete.

Hack 2 — Final Version: Complete & Polish POC v1.0

Date : 7 November • Saturday • Half-day

Aim : Use the final half-day to turn the improved POC into a complete, credible and demonstrable technical solution.

Activities

- Fully focused team build:
 - Fix remaining high-priority technical issues.
 - Polish outputs, reliability and demonstration flow.
 - Run final tests and capture evidence.
 - Prepare the project for formal presentation.

End-of-Hackathon deliverables

- **TECHNICAL DESIGN DOCS (Final Version)**
 - Final technical results / evaluation evidence.
- **FULLY BUILT POC VER 1.0** with all code and documentation pushed to **GitHub**.
 - Stable end-to-end technical demonstration.

CHECKPOINT: Complete the POC. Make it convincing.

Sprint 5 — Endgame: Package, Communicate & Defend

Dates : 8–14 November

Aim : Turn the completed POC into a clear, defensible project story and final submission.

Activities

- Complete final technical fixes from Hack 2.
- Clean and document the codebase.
- Prepare slides, video and poster.
- Connect problem → solution → technology → results.
- Prepare for technical questions and project defense.

End-of-Sprint deliverables

- **REVIEW WITH NGOs (table discussions).**
- Final POC VER 1.0 code repository with all final technical documentation pushed to GitHub.
- Ensure that the final versions of the following documents are on GitHub:
 - FUNCTIONAL SCOPE DOC
 - DESIGN DOC
 - TECHNICAL DESIGN DOC
 - Presentation Slides
 - Poster
 - Demo video / supporting media.
- Submit on Moodle
 - FINAL TECHNICAL DESIGN DOC with evaluation evidence.
 - Presentation slides.
 - Poster.
 - Demo video / supporting media.

CHECKPOINT: The project is complete, explainable and ready for final assessment.

End-of-Semester Evaluations

- 14 November — **Formal Demo 2:** Slides + video + code + poster + live demonstration.
- 18 November — **Final Assessment:** Table-wise assessment and project defense.
- 20 November — **Showcase:** Public exhibition in the KRB corridor.